

“We are looking at the The Linux Foundation solution set, ORAN ecosystems, open stack and Kubernetes solutions. There is also the Open Networking Foundation (ONF), where there are a set of open source platforms that exist. We are also looking at how DLT can be leveraged for smart contracts as the business models and monetisation is also important,” Gnanapriya states.

Creating a revolution

Linux and Android are a part of our life already. According to experts, they are playing a key role in the next phase of revolutionising the data centre market. For instance, a lot of infrastructure for hyperscalers like Google and Facebook runs on open source.

“We may not be aware that there are companies selling commercial networking products using open source to the data centre and cloud market. The next phase of the data centre evolution is the orchestration with OpenStack and Kubernetes,” says Abhijeet Chaudhary, founder and CEO, Niral Networks.

Acknowledging the pivotal role of 5G, IoT, edge and telecom, Chaudhary says there are multiple initiatives for open source disaggregation happening in 5G for RAN, core and transport. “There have been few smaller commercial 5G deployments using open source, but I think 5-6 years down the line, there will be a huge momentum towards open source based commercialisation in the telecom and private cellular networks for industrial/IoT use cases,” he adds.

Disaggregating components

When 5G came into existence, the thought process of disaggregating the components had already begun. With the arrival of open RAN and the disaggregated components talking to each other on open interfaces, the possibility of a lot of telecom applications in the RAN and transporting domains to become smaller components (that could be handled in isolation), opened up.



Abhijeet Chaudhary, founder and CEO, Niral Networks



Manish Gangey, SVP and head, R&D, Airtel



Dr Inder S. Gopal, CEO, Indian Urban Data Exchange

“These are components that you can innovate upon and bring a lot of value into before you put them together to create a larger solution. So, from the service provider perspective, these are great developments,” says Manish Gangey, SVP and head, R&D, Airtel.

He points out that one of the big problems in India is that we have a low ARPU (average revenue per user) in comparison to any other place. “So the way we need to build our own infrastructure has to be thought about differently for any company to be profitable.”

Open source fits beautifully in an association like this by reducing the overall cost of ownership, accelerating innovation and bringing in many more players into the market. “I look at open source as a great enabler for the Indian ecosystem to develop,” says Gangey.

Transforming industries

Dr Inder S. Gopal, CEO, Indian Urban Data Exchange, says there is evidence that open source can transform an industry. “If we look at data centre networking at the time of SDN and NFV, it was dominated by proprietary solutions. If we now look at the data centre and the boxes that are deployed in it, 80 per cent are white box solutions running, for the most part, open source software. The open source commercialisation has been driven by the availability of open white box hardware alternatives and mature open source offerings, and that has really transformed the data centre,” he explains.

He expects to see a similar kind of transformation in the telco space as well, though it might take a little bit longer. “I think telcos always move a little bit more slowly than other sectors, but it definitely will happen and there is evidence that it can happen,” Dr Gopal says.

Driving factor: Cost-effectiveness or competitiveness?

O-RAN based deployments are becoming increasingly popular. For an operator, the cost-effectiveness may be of prime importance, but there is also competition because of the low ARPU; hence, the infrastructure cost has to be matched in line with that. The question is: Are the cost reductions or the vendor lock-ins driving the popularity of open source? The fact is, reducing infrastructure costs will also bring in more competitors.